

Convexification

Non-Convex Problem:

$$v(p) = \min\{f(x; p) : g(x; p) \leq 0\}$$

Exact Convex Relaxation:

$$\tilde{v}(p) = v(p)$$

*Non-Convex
Solver
(Online)*



**Solution $\hat{x}$,
Cost $f(\hat{x}; p)$**

(No optimality guarantee)

*Convex
Solver
(Offline)*



**Neural Network:
 $\hat{v}(p) \approx \tilde{v}(p)$**

If $\hat{v}(p) \approx f(\hat{x}; p)$,
 $\hat{x}$ is near-globally optimal.

